

Number of girls who participate = 200.

$\therefore \frac{1}{2}$ of girls = 200 or total number of girls = 400.

Hence, total number of students = (300 + 400) = 700.

- 282.** Let the number of votes cast be x . Then, number of votes required = $\frac{3x}{4}$.

Counted votes = $\frac{2x}{3}$. Uncounted votes = $\left(x - \frac{2x}{3}\right) = \frac{x}{3}$.

Votes won by the candidate = $\frac{5}{6}$ of $\frac{3x}{4} = \frac{5x}{8}$.

Remaining votes required = $\left(\frac{3x}{4} - \frac{5x}{8}\right) = \frac{x}{8}$.

$\therefore$ Required fraction = $\frac{(x/8)}{(x/3)} = \left(\frac{x}{8} \times \frac{3}{x}\right) = \frac{3}{8}$.

- 283.** Let the total number of shots be x . Then,

Shots fired by A = $\frac{5}{8}x$; Shots fired by B = $\frac{3}{8}x$.

Killing shots by A = $\frac{1}{3}$ of $\frac{5}{8}x = \frac{5x}{24}$;

Shots missed by B = $\frac{1}{2}$ of $\frac{3}{8}x = \frac{3}{16}x$.

$\therefore \frac{3x}{16} = 27$ or $x = \left(\frac{27 \times 16}{3}\right) = 144$.

Birds killed by A = $\frac{5x}{24} = \left(\frac{5}{24} \times 144\right) = 30$.

- 284.** Number of alterations required in 1 shirt = $\left(\frac{2}{3} + \frac{3}{4} + \frac{4}{5}\right)$
 $= \frac{133}{60}$.

$\therefore$ Number of alterations required in 60 shirts = $\left(\frac{133}{60} \times 60\right)$
 $= 133$.

- 285.** Let the largest fraction be x and the smallest be y .

Then, $\frac{x}{y} = \frac{7}{6}$ or $y = \frac{6}{7}x$.

Let the middle one be z .

Then, $x + \frac{6}{7}x + z = \frac{59}{24}$ or $z = \left(\frac{59}{24} - \frac{13x}{7}\right)$.

$\therefore \frac{59}{24} - \frac{13x}{7} + \frac{1}{3} = \frac{7}{6} \Leftrightarrow \frac{13x}{7} = \frac{59}{24} + \frac{1}{3} - \frac{7}{6} = \frac{39}{24}$

$\Leftrightarrow x = \left(\frac{39}{24} \times \frac{7}{13}\right) = \frac{7}{8}$.

So, $x = \frac{7}{8}$, $y = \frac{6}{7} \times \frac{7}{8} = \frac{3}{4}$ and $z = \frac{59}{24} - \frac{13}{7} \times \frac{7}{8} = \frac{20}{24} = \frac{5}{6}$.

Hence, the fractions are $\frac{7}{8}$, $\frac{5}{6}$ and $\frac{3}{4}$.

- 286.** Let the number of hats purchased be x . Then, number of brown hats = $\frac{x}{4}$.

Number of hats sold = $\frac{2x}{3}$. Number of hats left unsold

$= \left(x - \frac{2x}{3}\right) = \frac{x}{3}$.

Number of brown hats sold = $\frac{4}{5}$ of $\frac{x}{4} = \frac{x}{5}$. Number of

brown hats left unsold = $\left(\frac{x}{4} - \frac{x}{5}\right) = \frac{x}{20}$.

$\therefore$ Required fraction = $\frac{\left(\frac{x}{20}\right)}{\left(\frac{x}{3}\right)} = \frac{x}{20} \times \frac{3}{x} = \frac{3}{20}$.

- 287.** Let the capacities of the bigger and smaller jars be x litres and y litres respectively.

Then, $\frac{x}{4} = \frac{y}{3} \Rightarrow y = \frac{3}{4}x$.

Quantity of water in bigger jar = $\left(\frac{x}{4} + \frac{y}{3}\right)$ litres

$= \left(\frac{x}{4} + \frac{1}{3} \times \frac{3}{4}x\right)$ litres = $\frac{x}{2}$ litres.

Hence, $\frac{1}{2}$ of the larger jar is filled with water.

- 288.** Let Mr. X's total income be ₹ x . Then, expenditure on food = ₹ $\frac{x}{4}$.

Expenditure on education = ₹ $\left[\left(1 - \frac{1}{3}\right)\frac{x}{4}\right]$
 $= ₹ \left(\frac{2}{3} \times \frac{x}{4}\right) = ₹ \frac{x}{6}$.

$\therefore$ Required fraction = $\frac{\left(\frac{x}{4} + \frac{x}{6}\right)}{x} = \frac{5x}{12} \times \frac{1}{x} = \frac{5}{12}$.

- 289.** Let the number of men in the 1st, 2nd, 3rd and 4th battalions be x , y , z and t respectively.

Then, $\frac{1}{2}x = \frac{2}{3}y = \frac{3}{4}z = \frac{4}{5}t \Rightarrow x = \frac{4}{3}y, z = \frac{8}{9}y, t = \frac{5}{6}y$.

Now, $x + y + z + t = 7300$

$\Rightarrow \frac{4}{3}y + y + \frac{8}{9}y + \frac{5}{6}y = 7300$

$\Rightarrow \frac{24y + 18y + 16y + 15y}{18} = 7300$

$\Rightarrow 73y = 7300 \times 18 \Rightarrow y = 1800$.

- 290.** Required fraction = Fraction of work left after Tuesday

$= 1 - \left(\frac{3}{5} + \frac{1}{3} \text{ of } \frac{2}{5}\right) = 1 - \left(\frac{3}{5} + \frac{2}{15}\right)$

$= 1 - \frac{11}{15} = \frac{4}{15}$.

- 291.** Let the total score be x . Then, highest score = $\frac{3x}{11}$.

Remainder = $\left(x - \frac{3x}{11}\right) = \frac{8x}{11}$. Next highest score